Computer Organization
Homework 3
(Due on 2018/11/12)

Note: Please hand in the hardcopy before the deadline.

1. Assume that the registers contain the following contents.
$t0=0x12345678, $t1=0xc3c3c3c3

(a) What are the values of $t2 and $t3 with the following sequence of instructions?(10%)

```
srl  $t2, $t1, 3
and $t2, $t2, $t0
ori $t3, $t2, -100
```

(b) Find the shortest sequence of MIPS instructions that extracts bits 20 down to 10 from register $t0 and put them into bits 10 down to 0 of register $t5 while the remaining bits are all zeros. (10%)

2. Translate the following C code to MIPS assembly code. Use a minimum number of instructions. Assume that the values of a, b, i, and j are in the register $s0, $s1, $t0, and $t1, respectively. Also assume that register $s2 holds the base address of the array U. (20%)

```
for (i=b; i<a; i++)
    for (j=i, j>=0, j--)
      U[i]=a-4*j
```

3. Assume $t0 holds the value 0xffffffcc. What is the value of $t2 after the following instructions? (10%)

```
        slti   $t1, $t0, -78
        bne    $t1, $zero, ELSE
        ori    $t2, $t0, 12
          j      DONE
ELSE:   sll    $t2,$t0,2
DONE:
```

4. Suppose that the program counter (PC) is set to 0x1000 0000.

(a) Is it possible to use the jump (j) MIPS assembly instruction to set the PC to the address as 0x1a00 0000? Please explain the reason. (5%)

(b) Is it possible to use to use the branch-on-equal (beq) MIPS assembly

instruction to set the PC to 0x0fff a000? Please explain the reason. (5%)

5. Translate the following loop into C. Assume that the C-level integer ***i*** is held in register $t1, $s1 holds the C-level integer called ***MyCounter,*** and $s2 holds the base address of the array ***YourArray.*** (20%)

```
        addi   $s1, $zero, 1
        sll    $t1, $s1, 2
LOOP:   add    $t3, $s2, $t1
        lw     $t4, 0($t3)
        sll    $t2, $t4, 4
        add    $s1, $s1, $t2
        addi   $t1, $t1, 4
        slti   $t2, $t1, 100
        bne    $t2, $zero, LOOP
```

6. Fill the blank (marked by yellow) in the table if it contains the machine codes for the following MIPS assembly instructions. (You can express in decimal values.) (20%)

```
LOOP1:  sll   $s3,$t1,3
        slt   $s2, $s3, $t2
        beq   $s2, $zero, EXIT
        sub   $s6, $s6, $t1
        sw    $s5, 24($s3)
        addi  $t1, $t1, 8
        j     LOOP1
EXIT:
```

Address	OP					
28244	0	0	(1)	(2)	(3)	0 (sll)
28248	0	(4)	10	(5)	0	42 (slt)
28252	4 (beq)	18	0	(6)		
28256	0	22	9	22	0	34 (sub)
28260	43(sw)	19	21	(7)		
28264	8 (addi)	(8)	(8)	(9)		
28268	2 (j)	(10)				
28272						